

# Final Internship Report

*A summary of an 8-week Web Development Internship*

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**Program:** CoderAxo Web Development Internship

**Offer ID:** CAX-OL-2026-310

**Internship Duration:** June 15, 2026 – August 15, 2026

**Report Submission Date:** August 14, 2026

## 1. Introduction

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This report summarises the eight weeks I spent as a Web Development Intern at CoderAxo, from June 15 to August 15, 2026. I joined the program to build a proper, structured foundation in web development — something beyond what a single university course usually covers — and to get hands-on experience with the same MERN stack (MongoDB, Express.js, React, and Node.js) that most real-world web teams use today. The internship was delivered through live Zoom sessions led by CoderAxo mentors, followed by independent practice, weekly activity logs, progress reports, and a series of hands-on tasks that gradually increased in difficulty, ending with a full-stack final project.

What I liked about the structure from the start was that it wasn't just lectures. Every topic was immediately followed by something to build, which meant the concepts didn't stay theoretical for long. By the end of the program, that approach took me from writing my first semantic HTML tags in Week 1 to shipping a complete MERN-stack team collaboration application in Week 8.

## 2. About CoderAxo

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CoderAxo is a web development training and internship program that runs cohort-based internships focused on practical, project-driven learning. The Web Development track I was part of is structured around the core technologies used to build modern full-stack applications: HTML, CSS, and JavaScript on the front end, Node.js and Express on the back end, React for building interactive user interfaces, and MongoDB as the database layer. Alongside the technical sessions, the program places a lot of weight on documentation, deadlines, and submission discipline — every task required a clean repository, a written report, and, for the final project, a recorded demo — which mirrors how work is actually reviewed in a professional setting.

## 3. Internship Objectives

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Going into the internship, my goals were fairly specific:

- Build a solid, practical foundation in HTML, CSS, and JavaScript rather than just a theoretical one
- Get comfortable with backend development using Node.js and Express
- Learn React well enough to build real, multi-page, API-connected applications

- Work through a complete MERN-stack project from planning to deployment, the way it would be done on an actual team
- Improve my ability to document, structure, and present my own work professionally

By the end of the eight weeks, I feel that all five of these were genuinely met, with the final project being the clearest proof of that — it required every one of those skills at once, under a real deadline.

## 4. The Learning Journey

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The program was structured in a clear progression, moving from front-end fundamentals to a complete full-stack build. I've broken this down by phase below rather than strictly by week, since several weeks flowed directly into each other.

### 4.1 Front-End Foundations (Weeks 1–3)

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The first three weeks covered HTML, CSS, and JavaScript, one per week, each through live sessions followed by practice. Week 1 focused on HTML5 — semantic structure, forms, tables, and writing markup that's actually meaningful rather than a stack of generic divs. Week 2 moved into CSS: the box model, Flexbox, Grid, positioning, and responsive design with media queries. I spent most of that week restyling the previous week's HTML pages, which was a good way to see immediately how layout decisions affect real content.

Week 3 introduced JavaScript — variables, functions, arrow functions, DOM manipulation, event handling, and working with arrays and objects. This is also when the first hands-on task, Task 1, was assigned: build a personal portfolio website using nothing but HTML, CSS, and JavaScript. I planned out an About, Skills, Projects, and Contact section, gathered content and images, and built the site with a JavaScript-powered mobile navigation toggle and a scroll-based active link highlight. It was the first time in the program that everything from the previous three weeks had to come together in one working project, and it made the connections between HTML, CSS, and JS click in a way that separate exercises hadn't.

### 4.2 Strengthening JavaScript & Introduction to the Backend (Weeks 4–5)

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Week 4 was built entirely around Task 2 — a To-Do application with task creation, deletion, a 'mark as completed' toggle, a dark/light mode switch, and data persistence using localStorage. This task pushed my JavaScript further than the portfolio had, particularly around keeping the UI, the underlying array of tasks, and localStorage all in sync every time something changed. Re-rendering the task list on every update turned out to be the simplest and most reliable fix, and it's a pattern I've reused since.

Week 5 shifted to the backend with an introduction to Node.js. This started with the core ideas — how Node lets JavaScript run outside the browser, how npm and package.json manage dependencies — and moved into building a very small HTTP server from scratch using Node's built-in http module before being introduced to Express.js. Learning the raw version first made Express's routing and middleware make a lot more sense once we got to it. By the end of the week I was comfortable writing basic GET/POST routes and returning JSON responses, which set up the backend half of everything that came after.

### 4.3 React and Task 3 (Weeks 6–7)

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Week 6 introduced React: components, JSX, props, state via `useState`, side effects via `useEffect`, and basic routing with React Router. It was also the week the Final Project brief landed, on July 25 — the Project & Team Collaboration System, to be built on the mandatory MERN stack within a 10-day window. A good part of the week went into planning: reading through the full requirements, sketching a database schema, and breaking the project into modules before writing any code.

Week 7 ran two things in parallel. The first was Task 3, which asked for a complete, deployed React application with functional routing, real API data, and proper styling. I built a countries-explorer app with a Home page, individual country detail pages, an About page, a Contact page, and a 404/error page, all connected through React Router, styled with Tailwind CSS, and deployed publicly. The second was the actual start of Final Project development. I used a vertical-slice approach — building the backend and frontend for one module together before moving to the next — and started with Authentication: an Express and MongoDB Atlas backend with JWT-based login and `bcrypt` password hashing, paired with a React/Vite frontend using an `AuthContext` and a protected Dashboard route. Getting Auth working end to end first gave me a stable foundation to build every other module on top of.

### 4.4 Final Project Completion (Week 8)

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The last full week was almost entirely focused on finishing the Final Project. This is covered in detail in Section 5, but broadly it involved building out the Dashboard, Team Management, Project CRUD, the Kanban-style Task Board, File Sharing, and Notifications modules, layering in security throughout, and then testing, documenting, and submitting the completed application ahead of the August 5 deadline.

## 5. Final Project: Project & Team Collaboration System

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The Final Project was the centrepiece of the internship and, realistically, the part I'm most proud of. It's a full-stack MERN application built to help teams plan, assign, and track work in a single dashboard, and it had to be built and submitted within 10 days of the brief being released.

### 5.1 Technology Stack

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- Frontend: React 18, Tailwind CSS, Framer Motion, Recharts, Axios, Lucide Icons
- Backend: Node.js, Express.js, JWT Authentication, Bcrypt.js
- Database: MongoDB with Mongoose ODM

### 5.2 Core Modules

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- Authentication & Account Management — secure registration and login with JWT and `bcrypt`-hashed passwords, plus profile settings
- Dashboard — real-time summary cards, task-status distribution, and a recent-activity feed
- Team Management — creating teams, adding members, and assigning roles

- Project Management — full CRUD for projects with status tracking (planning, in-progress, completed)
- Task Board (Kanban) — tasks organised into Todo, In Progress, Review, and Completed, with priority levels and due dates
- File Sharing — a dedicated module for uploading and managing project documents
- Notifications — activity-based alerts for team and task updates
- Security — input validation, protected routes, authorization checks, and safe password storage throughout

On the database side, MongoDB collections were normalised and linked through ObjectId references: a Team can have multiple members and projects; each Project holds multiple Tasks assigned to individual users; and Activities and Notifications reference back to the relevant users, projects, or tasks. Building it in a vertical-slice order — Auth, then Dashboard, then Teams and Projects, then the Task Board, then File Sharing and Notifications — meant there was always a working application at every stage of the 10 days, rather than a pile of disconnected pieces coming together only at the very end.

For submission, the requirements were strict: complete source code, a database export, a GitHub repository link, a README with installation instructions, screenshots, an ER diagram, API documentation, and a 5–10 minute demo video, all referenced through a Google Drive link inside a written Final Project Report. I completed all of these — wrote the README and API documentation, generated the ER diagram, recorded the demo, and compiled the report — and submitted the project ahead of the August 5 deadline.

## 6. Technical Skills Gained

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- Semantic, accessible HTML5 and responsive CSS using Flexbox and Grid together
- Core JavaScript — DOM manipulation, event handling, arrays/objects, and ES6 syntax such as arrow functions
- Client-side data persistence with localStorage and JSON
- Server-side fundamentals with Node.js and REST API design using Express.js
- Building component-based, stateful interfaces in React, including hooks (useState, useEffect) and React Router
- Consuming real APIs in React and handling loading/error states properly
- Full MERN-stack integration — connecting a React frontend to an Express/MongoDB backend end to end
- Authentication and security fundamentals — JWT, password hashing with bcrypt, protected routes, and input validation
- Using Git and GitHub for version control, and deploying front-end applications publicly

## 7. Soft Skills and Professional Growth

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Beyond the technical stack, this internship pushed skills that don't show up directly in code but matter just as much. Working against real weekly deadlines — activity logs, progress reports, and task submissions every single week — built a level of time discipline that's hard to get from self-paced learning alone. The Final Project in particular, with its hard 10-day window, forced me to plan realistically, prioritise which features mattered most, and accept that not everything could be polished equally in the time available.

Writing documentation — READMEs, API docs, progress reports — also turned out to be its own skill. It's one thing to know how a project works because you built it; it's another to explain it clearly enough that someone who has never seen the code could set it up and understand it. I got noticeably more comfortable with that over the course of the program.

## 8. Challenges Faced and How I Overcame Them

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A few specific challenges stood out. Early on, a Flexbox alignment bug in Week 2 — a navbar not centering vertically — turned out to be caused by a missing height on the parent container, a small mistake that took longer than it should have to spot. In Week 4, keeping the To-Do App's task list synced correctly across localStorage, the underlying data array, and the UI took a few attempts before settling on simply re-rendering the whole list on every change, which was simpler and more reliable than trying to patch individual DOM nodes.

The bigger challenge was pacing during the Final Project. Ten days is not a lot of time for a full MERN application with eight modules, authentication, and security requirements, especially while Task 3 was still due in the same stretch. The vertical-slice approach — finishing Authentication completely before moving on, rather than half-building every module at once — was what made the deadline realistic. It meant that even if I had run out of time, there would have been a working, demonstrable product rather than a set of unfinished pieces.

## 9. Mentorship and Support

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CoderAxo's live Zoom sessions were genuinely useful rather than just a formality — each one covered the week's topic in enough depth to actually build something with it afterward, and questions were addressed directly rather than deferred. Having a structured weekly rhythm of session, practice, and submission, with a mentor available to check in on progress, made it much easier to stay consistent over eight weeks than self-study usually is.

## 10. Conclusion

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This internship took me from writing my first lines of semantic HTML to shipping a complete, secured, full-stack MERN application in eight weeks, and the structure — weekly sessions, hands-on tasks, and a real final project with a real deadline — is exactly what made that possible. The Project & Team Collaboration System is the piece of work from this internship I'd point to first when showing what I can build, and the process of getting there — Authentication, then Dashboard, then every module after it, documented and shipped on time — is a workflow I plan to keep using going forward.

Going into this, my longer-term goal has been to move toward a Java Developer role while keeping my web development skills sharp on the side through freelancing. This internship strengthened the JavaScript, React, and backend fundamentals that carry over directly into that goal, and gave me a complete project I can reference in interviews and on my portfolio. I'm grateful to the CoderAxo team for the structure, the mentorship, and the push to build something real rather than just learn in theory, and I'm looking to carry the habits from these eight weeks — planning before building, documenting as I go, and shipping in working slices — into whatever I build next.